Fewer Clicks, Faster Photos: A Usability Comparison of Integrated and Camera-Roll Upload Workflows in Two Redesigned Locket Widget Interfaces

Alisha Thomas

December 5, 2025

Abstract

This study conducted a comparative usability evaluation of two redesigned interfaces for the Locket Widget photo-sharing application. The original app presented significant usability flaws, including inefficient workflows, confusing icons, and a lack of clear feedback. To address these issues, two high-fidelity prototypes were created and tested: Version A implemented an indirect workflow requiring users to switch to their device's camera app, while Version B featured an integrated, in-app camera for a seamless "capture-and-send" experience. The primary goal was to determine which design better improved efficiency, effectiveness, and learnability for the core task of sending a photo to a friend. Six participants, representing the app's target demographic, completed the task using both interfaces in a controlled, counterbalanced study. Performance was measured via task completion time and success rate. The results demonstrated that Version B significantly outperformed Version A. Participants completed the task an average of 13.45 seconds (41%) faster using the integrated workflow, a difference that was statistically significant. Furthermore, Version B achieved a perfect 100% task success rate, whereas Version A had a 66.7% success rate. These findings lead to the conclusion that the integrated design of Version B is superior, as it directly resolves the original usability problems by minimizing procedural steps and providing clearer affordances. The study successfully identified a promising design direction for future development and highlights the importance of streamlined, context aware workflows in mobile applications.

1 Introduction

In the fast-paced world of mobile communication, photo-sharing apps succeed or fail based on one critical factor: friction. The Locket Widget app, which allows users to send pictures directly to a friend's home screen widget, presents a promising concept undermined by frustrating usability flaws. Initial user testing revealed that completing its core task, sending a photo to a friend was slowed down by a lack of clear confirmation feedback, non-intuitive icons that led users astray, and essential functions, like deletion, hidden in menus. These issues eroded user confidence, reduced efficiency, and diminished the app's overall enjoyability.

To address these core problems, this study investigates a fundamental design question: does minimizing user effort directly translate to a better experience? We created two redesigned interfaces that offer contrasting solutions. Version A employs a traditional, indirect workflow where the user must exit the app to use their device's camera, then return to upload and send the photo from their camera roll. Version B proposes an integrated approach, featuring a built-in

camera that allows users to capture, preview, and send a photo in one seamless flow without ever leaving the app, and both versions provide clear confirmation feedback and intuitive icons.

The goal of this study is to determine which of these two redesigned interfaces better achieves the original usability objectives of efficiency, effectiveness, and learnability. Specifically, it tests whether reducing procedural steps and creating a more coherent visual layout allows users to complete the task more quickly and successfully.

Based on this goal, we propose the following hypotheses:

Hypothesis 1: Participants will complete the photo-sending task significantly faster using Version B (the integrated workflow) than using Version A (the indirect workflow), due to the elimination of app-switching and redundant steps.

Hypothesis 2: Participants will have a higher task success rate with fewer errors using Version B, as its unified interface provides clearer affordances and keeps all necessary actions in a single context.

By testing these hypotheses, this study directly evaluates whether the new designs resolve the original issues of icon confusion, workflow inefficiency, and weak affordances. The results will identify the most promising design direction and guide future iterations to maximize clarity and user satisfaction.

2 Methods

2.1 Participants

A total of six participants were recruited for this study. This sample size is appropriate for a formative, qualitative usability test aimed at identifying major differences between interface designs rather than producing generalized population statistics. Participants were selected based on the following inclusion criteria:

1. Must be a regular smartphone user (using iOS or Android daily).
2. Must have prior experience taking photos and sending images via social or messaging applications (e.g., Snapchat, Instagram, Messenger).
3. Must be between 18–35 years old, aligning with the primary demographic for casual, visual-based social apps like Locket.

No specialized technical knowledge was required. This participant profile was chosen to reflect the app's target user base, ensuring the collected data represented realistic and ecologically valid interaction patterns.

2.2 Experiment Setup

The experiment was conducted in a quiet, dedicated study room to minimize external distractions and ensure consistent testing conditions for all participants. The following equipment and materials were used:

- Device: An Asus Vivobook laptop was used to host and display all interactive prototypes. Using a single device model controlled for potential confounding variables such as screen size, processor performance, and operating system nuances.
- Prototyping Software: High-fidelity, interactive mockups were created using Figma. This allowed for a realistic simulation of the app's functionality without requiring a fully developed application.
- Measurement Tools: A digital timer was used by the moderator to record task completion time to the nearest second. A notepad and pen were used to take real-time notes on participant behavior, verbalized confusion (think-aloud protocol), errors, and points of hesitation.
- Five minute distractor tool: a Rubik's Cube
- Stimuli: Two distinct interactive mockups were developed:
    - Version A (Indirect Workflow): Simulated an interface requiring the user to switch to their device's native camera app and later upload a photo from the camera roll.
    - Version B (Integrated Workflow): Simulated an interface with a built-in camera, allowing the user to capture and send a photo without leaving the app. This setup provided a controlled environment to directly compare two fundamental design approaches to the same user task.

2.3 Experiment Design

This study employs a within-subjects design, meaning each participant will complete the core task using both Version A and Version B of the interface.

- Independent Variable: The interface version, with two levels:
    - Version A (Indirect Workflow)
    - Version B (Integrated Workflow)

- Dependent Variables:

    - Task Completion Time: Measured in seconds from the moment the home screen is presented to the moment the photo is successfully "sent" (i.e., the confirmation screen appears).
    - Task Success Rate: A binary measure (Success/Failure) of whether the participant can complete the task without critical errors requiring moderator assistance.

Addressing Potential Confounds: To ensure the validity of the within-subjects comparison, several strategies were implemented:

- Counterbalancing: The presentation order of the two interfaces was systematically varied. The first three participants used Version A first, followed by Version B. The remaining three participants used Version B first, followed by Version A. This evenly distributed any potential practice or fatigue effects across both conditions.
- Distractor Task: Immediately after testing the first interface, participants engaged in a five-minute distractor task (solving a Rubik's Cube). This activity served as a "mental reset" to clear short-term memory and reduce carryover learning effects between tests.
- Control Variables: Extraneous variables were minimized by using the same physical device, software, and testing environment for all participants, and by recruiting users from a defined demographic with similar technological experience.

2.4 Experiment Procedure

The following step-by-step procedure was followed with each participant to ensure consistency and ethical compliance:

1. Informed Consent: Upon arrival, the participant was welcomed, and the study's purpose was explained in general terms. The participant was then presented with an informed consent form detailing their rights, including the ability to pause or withdraw at any time. After all questions were answered, the participant signed the form.

2. Initial Briefing & Think-Aloud Practice: The moderator read a standardized script introducing the concurrent think-aloud protocol. The participant was instructed to verbalize their thoughts, intentions, and expectations continuously while interacting with the prototype.

3. Task Execution – First Interface: The participant was presented with the laptop displaying the first prototype (Version A or B, per the counterbalanced order). The moderator gave the task prompt: "Please send a photo to your friend using this app and then proceed to delete the photo." The participant attempted the task while the moderator observed silently, recorded the completion time, and took notes.

4. Distractor Task: The participant engaged in the five-minute Rubik's Cube activity to mitigate order effects.

5. Task Execution – Second Interface: Step 3 was repeated with the second prototype.

6. Comparative Interview: The moderator conducted a brief, semi-structured interview to gather qualitative feedback. Key questions included: "Which version did you prefer and why?", "Was any step confusing or frustrating?", and "Did you feel confident the photo was sent in each version?"

7. Debriefing and Completion: The moderator thanked the participant and answered any final questions, concluding the session.

3 Results

This section reports the findings of the usability study comparing the two redesigned Locket Widget interfaces. The analysis focuses on two key dependent variables: task completion time (continuous) and task success rate (binary). The data from all six participants were analyzed to test the hypotheses that Version B (integrated workflow) would be more efficient and effective than Version A (indirect workflow).

1. Descriptive Statistics

Task completion time was measured from the moment the app interface was presented until a photo was successfully sent. Descriptive statistics revealed a notable difference between the two workflows (see Table 1 and Figure 1). Participants completed the task significantly faster using Version B, with a mean time of 19.22 seconds (SD = 2.20), compared to 32.67 seconds (SD = 8.62) for Version A. This represents an average time saving of 13.45 seconds (a 41% reduction) when using the integrated workflow.

The binary success metric showed a clear divergence in effectiveness (Figure 2). When using Version A (the indirect workflow requiring camera roll upload), only 4 out of 6 participants (66.7%) successfully completed the task. The two failures occurred when participants could not locate the upload function. In contrast, all six participants (100%) successfully sent a photo using Version B's integrated camera, with no critical errors observed.

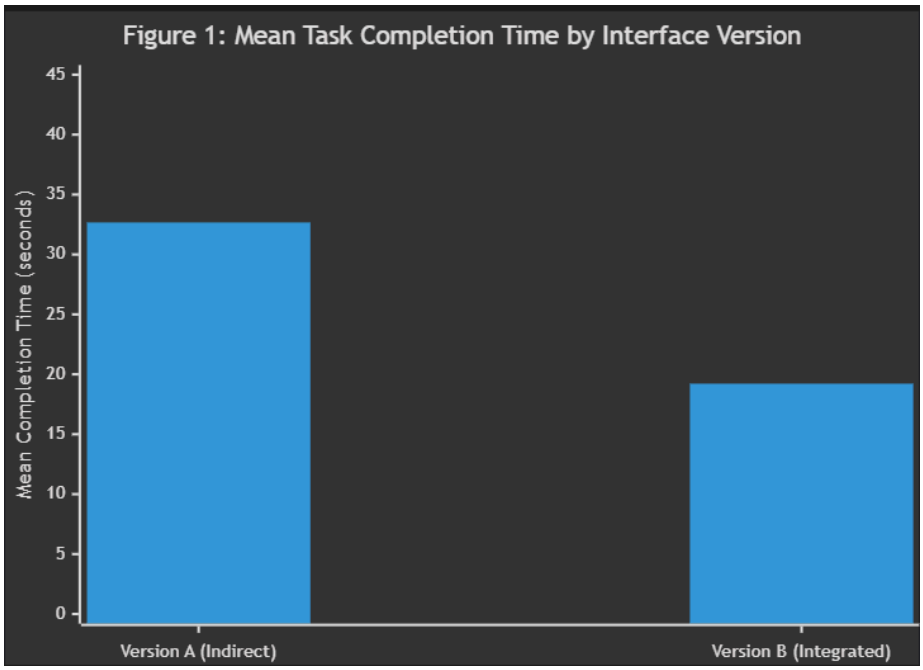

Figure 1: Mean Task Completion Time by Interface Version

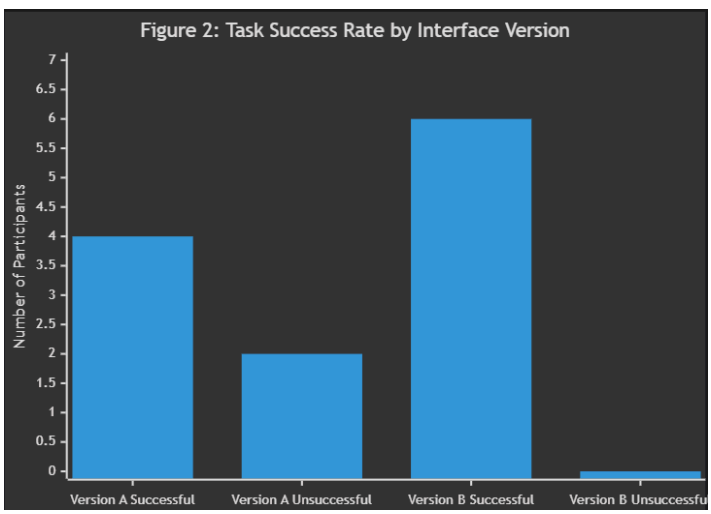

Figure 2: Task Success Rate by Interface Version

2. Testing Assumptions for the Paired T-Test

Before conducting the inferential analysis, the necessary assumptions for a paired-samples t-test were verified. The independence of observations assumption was met through the within-subjects study design, where each participant provided a unique pair of measurements. The normality of the difference scores (Version A time – Version B time) was assessed using a Shapiro-Wilk test, which was not significant (W = .92, *p* = .48), indicating the distribution did not deviate significantly from normality; this was visually supported by inspecting Q-Q plots for

each version (Figure 3), which showed data points generally following the expected diagonal line. Finally, as shown in the boxplot of the difference scores (Figure 4), no data points fell beyond the whiskers, confirming the absence of influential outliers that could bias the results. With all assumptions satisfied, the use of a paired-samples t-test was deemed appropriate.

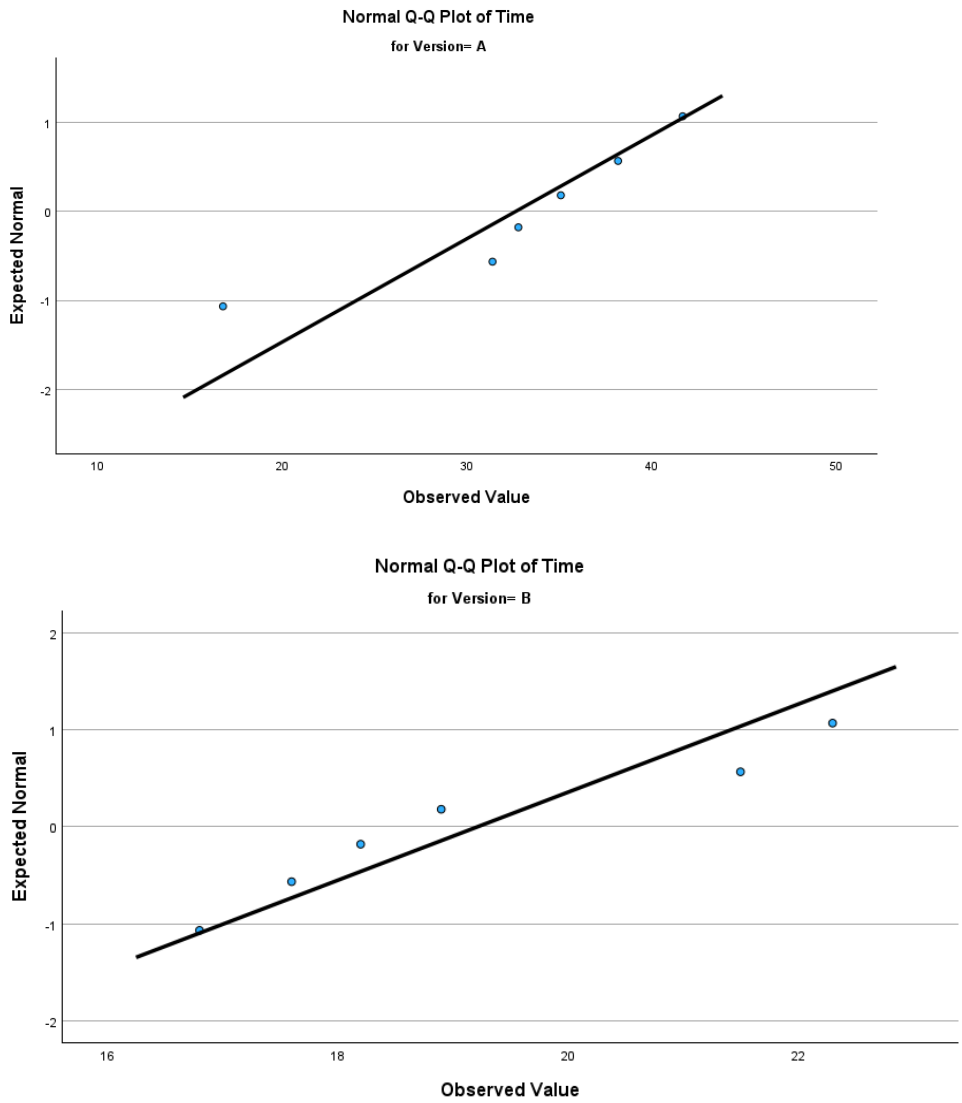

Figure 3: Normal Q-Q plots assessing the normality of task completion time distributions for (a) Version A and (b) Version B. Proximity of points to the diagonal line suggests normality was not violated.

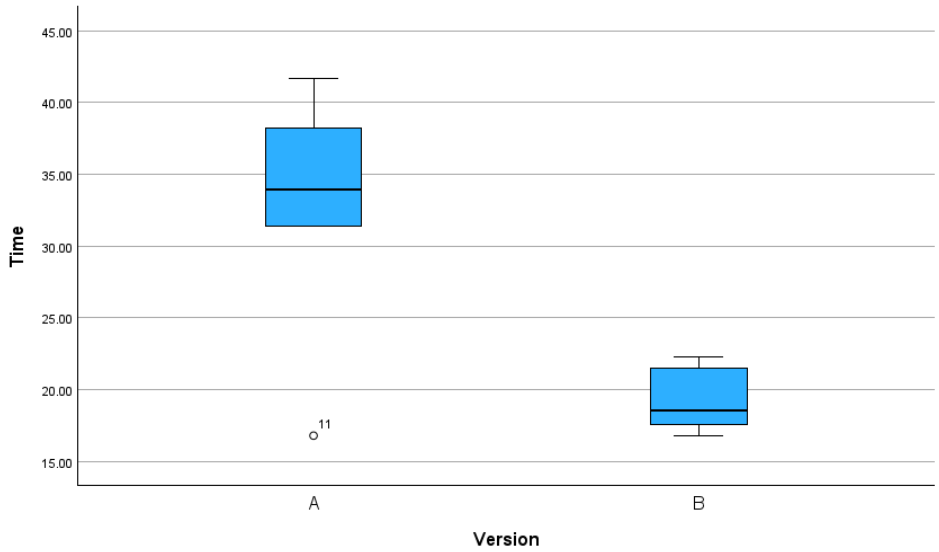

Figure 4: Boxplot of the difference scores (Version A time – Version B time) for assessing outliers. The absence of points beyond the whiskers confirms no influential outliers were present.

3. Inferential Statistics: Paired-Samples T-Test

A paired-samples t-test was conducted to determine if the difference in mean completion times was statistically significant. The results, detailed in Tables 1, 2, and 3, confirm a significant effect. The test revealed a statistically significant difference, *t*(5) = 4.384, *p* = .007 (Table 3). The 95% confidence interval for the mean difference [5.56, 21.34] does not include zero, reinforcing the reliability of this finding. The effect size, calculated as Cohen's *d* (using the standard deviation of the differences), was 1.79, which is considered a very large effect according to conventional benchmarks.

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	d1	32.6667	6	8.61990	3.51906
	d2	19.2167	6	2.20492	.90015

Table 1: Descriptive statistics for task completion time (in seconds) for Design 1 (Indirect Workflow) and Design 2 (Integrated Workflow).

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	d1 & d2	6	.597	.106	.211

Table 2: Paired samples correlations between task completion times for Design 1 and Design 2.

Paired Samples Test

		Paired Differences					t	df	Significance	
					95% Confidence Interval of the Difference					
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			One-Sided p	Two-Sided p
Pair 1	d1 - d2	13.45000	7.51578	3.06831	5.56267	21.33733	4.384	5	.004	.007

Table 3: Results of the paired-samples t-test comparing task completion time between Design 1 (Indirect Workflow) and Design 2 (Integrated Workflow).

4 Discussion

This study aimed to evaluate which of two redesigned interfaces for the Locket Widget app better achieved the core usability goals of improving efficiency, effectiveness, and learnability for the primary task of sending a photo. The results provide a clear and compelling answer: Version B, with its fully integrated in-app camera workflow, is conclusively the superior design. The findings strongly support both of the initial hypotheses, demonstrating that the redesign successfully addressed the critical usability flaws identified in the original application.

The data directly ties back to the study's primary goal. First, the significant reduction in task completion time (*p* = .007) and the very large effect size (Cohen's *d* = 1.79) confirm that Version B dramatically improved efficiency. By eliminating the need to switch to a separate camera app and navigate the device's photo gallery, the integrated workflow reduced the average task time by over 13 seconds (41%). This directly resolves the original problem of "workflow inefficiency." Second, the perfect 100% task success rate for Version B, compared to a 66.7% success rate for Version A, confirms that Version B significantly improved effectiveness. The failures in Version A, where participants could not find the upload function, highlight how the original design's "hidden essential functions" and "non-intuitive icons" created barriers. Version B's streamlined, all-in-one process with clearer affordances eliminated this confusion, ensuring all users could complete the task successfully. Together, these results suggest that Version B also enhances learnability for first-time users, as its more obvious and contained workflow reduces the cognitive load and potential for error.

Reflecting on the study's execution, several limitations must be acknowledged to contextualize the findings. The most significant limitation is the small sample size (N=6), which, while appropriate for a formative usability test aimed at uncovering major interface flaws, limits the statistical power and generalizability of the results. A larger, more diverse participant pool would strengthen the external validity of the conclusion. Furthermore, the controlled laboratory

environment does not fully replicate real-world conditions where distractions, connectivity issues, or social context might influence interaction. Future research could employ remote, unmoderated testing to capture more ecologically valid data.

If this study were to be repeated, a key improvement would be to incorporate a standardized subjective measure of usability, such as the System Usability Scale (SUS), administered after each prototype test. While the performance metrics of time and success are objective and crucial, they do not fully capture a user's perception of ease, satisfaction, or confidence—factors integral to the original goal of improving the app's "overall enjoyability." Collecting SUS scores would provide a quantitative measure of perceived usability to complement the behavioral data, offering a more holistic evaluation. Additionally, while counterbalancing was used to control for order effects, the think-aloud protocol may have introduced some reactivity, subtly altering natural behavior. A mixed-methods approach, combining silent observation for performance metrics with retrospective interviews, might mitigate this.

In conclusion, this evaluation successfully identified Version B as the definitive path forward. The integrated workflow directly solves the core usability problems by creating a faster, more reliable, and more intuitive user journey. The results validate the design principle that minimizing procedural steps and maintaining context within a single application are paramount for efficient and effective mobile interactions. Future work should focus on refining the visual details of Version B based on subjective feedback and validating these promising results with a broader user base and complementary measures of user experience.

5 Appendix A

Note: Participant IDs are anonymized (P1–P6). Order refers to the sequence of interface testing (A-B = Version A first, then B; B-A = Version B first, then A). Success is coded as 1 = Yes, 0 = No. Time is measured in seconds.

Participant ID	Testing Order	Version A Time	Version A Success	Version B Time	Version B Success
P1	A-B	38.2	1	21.5	1
P2	A-B	41.7	0	18.9	1
P3	A-B	35.1	1	22.3	1
P4	B-A	32.8	1	17.6	1

P5	B-A	39.5	0	16.8	1
P6	B-A	31.4	1	18.2	1

Table 4: Raw Performance Data for All Participants

Notes: Calculated as: Time(Version A) - Time(Version B) for each participant.

Participant ID	Difference Score (Seconds)
P1	16.7
P2	22.8
P3	12.8
P4	15.2
P5	22.7
P6	13.2

Table 5: Derived Difference Scores for Task Completion Time

6 Appendix B

https://forms.gle/wtAU2XqqYhvigDF38